

1 MINIMUM ZONE

Bars	#5 minimum, #11 maximum
Faces	provide the same rows each face of wall
Rows	3 rows minimum each face
Cover	1 in clear cover
Row spacing	6 in on center after first row

2 ROW LAYOUT

Straight edge	3 rows each face: 1 in clear to bar edge, then 6 in o.c.
Overlap corner	4 bars in the overlap box, 1 in clear from box/wall edges
Corner continuation	after the overlap box, offset row pairs at 6 in on center along each leg

3 PLACEMENT MATH

$$e_b = c_{clr} + \frac{d_b}{2} \quad (1)$$

e_b	bar centerline offset from concrete edge [in]
c_{clr}	clear cover from concrete edge to nearest bar edge, use 1 in
d_b	selected bar diameter [in]

$$r_k = e_b + (k - 1)s_r, \quad k = 1, 2, 3 \quad (2)$$

r_k	straight-edge row k centerline distance from wall edge [in]
k	row index counted from wall edge
s_r	row spacing, use 6 in on center

$$(x, y) \in \{(e_b, e_b), (b_x - e_b, e_b), (e_b, b_y - e_b), (b_x - e_b, b_y - e_b)\} \quad (3)$$

x	local bar centerline coordinate along overlap-box width [in]
y	local bar centerline coordinate along overlap-box depth [in]
b_x	overlap-box width measured along the first wall leg [in]
b_y	overlap-box depth measured along the second wall leg [in]

$$q_p = (b - e_b) + ps_r, \quad p = 1, 2 \quad (4)$$

q_p	continuation-row centerline distance from overlap-box origin [in]
b	overlap-box dimension along the continuing wall leg [in]
p	continuation-row index outside overlap box

#5 CENTER OFFSET

d_b	0.625 in
e_b	1.3125 in

4 RULE

- Put bond-zone rows at wall ends, wall corners, and opening jamb edges.
- At a straight edge, place 3 rows minimum on each wall face.
- At an overlapping corner, place 4 bars in the corner overlap box.
- Keep corner-box bars 1 in clear to the bar edge, not to the bar centerline.
- From the corner box, continue row pairs at 6 in on center along each leg.
- Increase bar size only as needed, from #5 up to #11 maximum.

5 EXAMPLES

Straight-end groups have 6 bars minimum. Overlap-corner groups share the first row in a 4-bar box.

U WALL: 12 FT BY 12 FT

Zone	Layout	Min bars
Left leg free end	3 rows each face	6
Left/back overlap corner	4-bar box + 2 row pairs each leg	12
Right/back overlap corner	4-bar box + 2 row pairs each leg	12
Right leg free end	3 rows each face	6
Total	2 free ends + 2 overlap corners	36

L WALL: 16 FT BY 8 FT

Zone	Layout	Min bars
Long leg free end	3 rows each face	6
Overlap corner	4-bar box + 2 row pairs each leg	12
Short leg free end	3 rows each face	6
Total	2 free ends + 1 overlap corner	24

T WALL: 15 FT WITH 8 FT MIDPOINT STEM

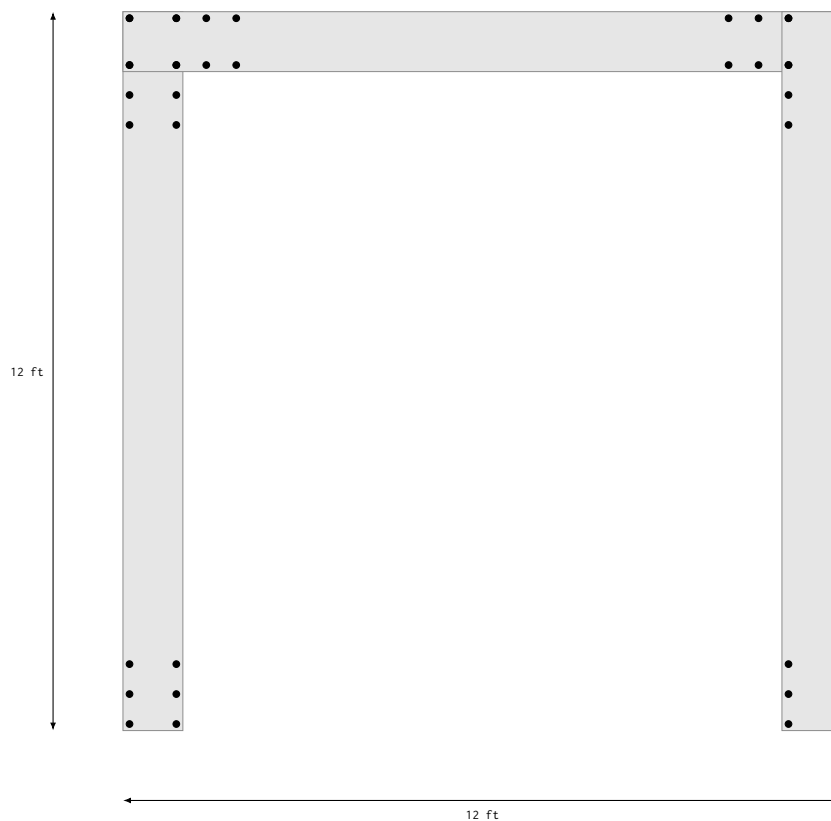
Zone	Layout	Min bars
Left end of 15 ft wall	3 rows each face	6
Right end of 15 ft wall	3 rows each face	6
Stem free end	3 rows each face	6
Midpoint overlap box	4-bar box + 2 row pairs each branch	16
Total	3 free ends + 1 three-branch overlap	34

6 CROSS SECTIONS

Dot one #5 minimum bar position
Gray wall schematic wall thickness; only cover and row spacing are being checked
Figure offset dot centers drawn at 1.3125 in from concrete edge for #5 bars

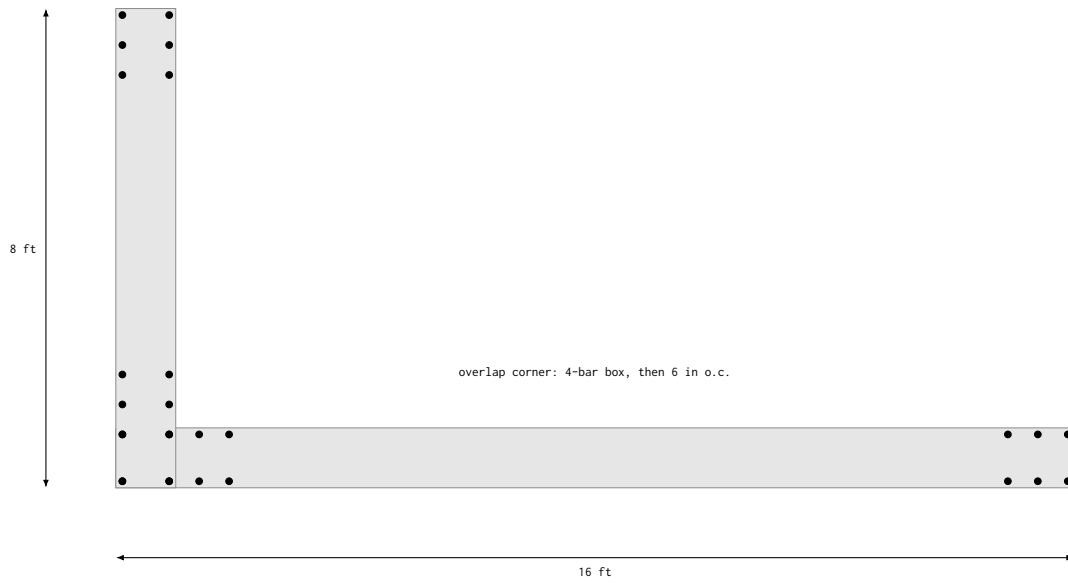
U WALL: 12 FT BY 12 FT

overlap corners: 4-bar box, then 6 in o.c.



2 free ends + 2 overlap corners = 36 minimum positions

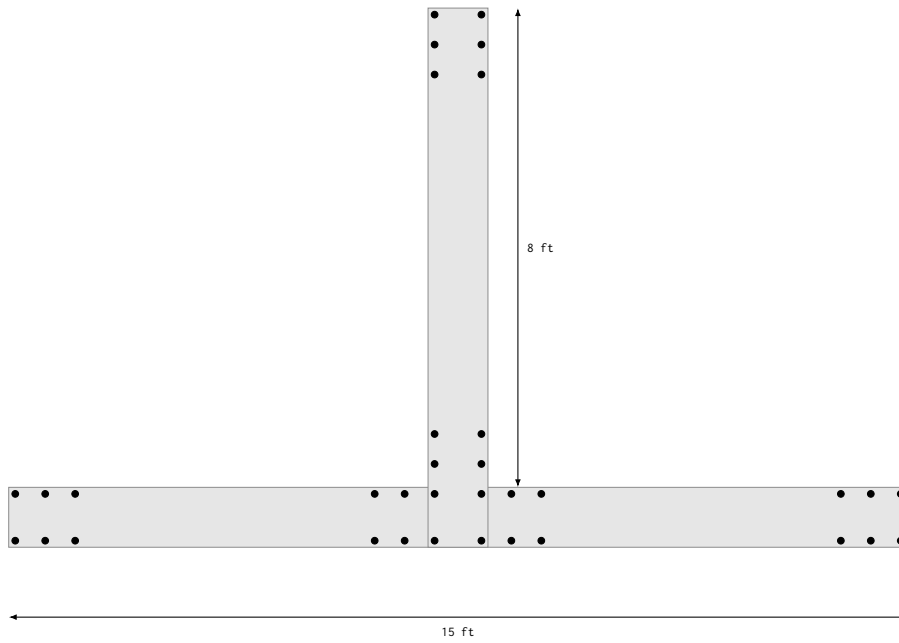
L WALL: 16 FT BY 8 FT



2 free ends + 1 overlap corner = 24 minimum positions

T WALL: 15 FT WITH 8 FT MIDPOINT STEM

midpoint overlap: 4-bar box, then 6 in o.c. each branch



3 free ends + midpoint overlap = 34 minimum positions